

Task F - Outlier Detection Report (Page 1 of 12)

Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	IQR Outliers	Z-Score Outliers
TransactionID	3134634.0	3429903.0	295269.0	2691730.5	3872806.5	0	0
isFraud	0.0	0.0	0.0	0.0	0.0	20663	20663
TransactionDT	3027036.0	11246585.0	8219549.0	-9302287.5	23575908.5	0	0
TransactionAmt	43.321	125.0	81.679	-79.1975	247.5185	66481	10093
card1	6019.0	14184.0	8165.0	-6228.5	26431.5	0	0
card2	215.0	512.0	297.0	-230.5	957.5	0	0
card3	150.0	150.0	0.0	150.0	150.0	67688	5752
card5	166.0	226.0	60.0	76.0	316.0	0	0
card6	0.0	1.0	1.0	-1.5	2.5	0	0
addr1	205.0	327.0	122.0	22.0	510.0	8807	0
addr2	87.0	87.0	0.0	87.0	87.0	4353	4338
dist1	8.0	8.0	0.0	8.0	8.0	230536	12986
P_emaildomain	0.0	2.0	2.0	-3.0	5.0	61714	19498
C1	1.0	3.0	2.0	-2.0	6.0	59534	2024
C2	1.0	3.0	2.0	-2.0	6.0	62213	2335
C3	0.0	0.0	0.0	0.0	0.0	2429	2429
C4	0.0	0.0	0.0	0.0	0.0	138657	2017
C5	0.0	1.0	1.0	-1.5	2.5	60444	19398
C6	1.0	2.0	1.0	-0.5	3.5	78328	2314
C7	0.0	0.0	0.0	0.0	0.0	67398	1277

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Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	IQR Outliers	Z-Score Outliers
C8	0.0	0.0	0.0	0.0	0.0	142873	2024
C9	0.0	2.0	2.0	-3.0	5.0	42628	19914
C10	0.0	0.0	0.0	0.0	0.0	137098	2051
C11	1.0	2.0	1.0	-0.5	3.5	74843	2041
C12	0.0	0.0	0.0	0.0	0.0	101340	1300
C13	1.0	12.0	11.0	-15.5	28.5	75630	19700
C14	1.0	2.0	1.0	-0.5	3.5	96089	3240
D1	0.0	121.0	121.0	-181.5	302.5	75153	11650
D2	88.0	109.0	21.0	56.5	140.5	248677	16310
D3	7.0	11.0	4.0	1.0	17.0	182512	12710
D4	0.0	123.0	123.0	-184.5	307.5	90852	5254
D5	10.0	10.0	0.0	10.0	10.0	276568	16373
D10	0.0	150.0	150.0	-225.0	375.0	74195	3557
D11	31.0	57.0	26.0	-8.0	96.0	128896	15757
D15	0.0	251.0	251.0	-376.5	627.5	6618	397
M1	0.0	0.0	0.0	0.0	0.0	25	25
M2	0.0	0.0	0.0	0.0	0.0	33972	33972
M3	0.0	0.0	0.0	0.0	0.0	67709	0
M5	0.0	0.0	0.0	0.0	0.0	107566	0
M6	0.0	1.0	1.0	-1.5	2.5	0	0

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Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	IQR Outliers	Z-Score Outliers
M7	0.0	0.0	0.0	0.0	0.0	32901	32901
M8	0.0	0.0	0.0	0.0	0.0	89037	0
M9	0.0	0.0	0.0	0.0	0.0	38631	38631
V1	1.0	1.0	0.0	1.0	1.0	17	17
V2	1.0	1.0	0.0	1.0	1.0	12359	12359
V3	1.0	1.0	0.0	1.0	1.0	20670	20670
V4	0.8465	1.0	0.1535	0.6161	1.2303	64752	8066
V5	0.877	1.0	0.123	0.6925	1.1845	68369	14094
V6	1.0	1.0	0.0	1.0	1.0	12626	12626
V7	1.0	1.0	0.0	1.0	1.0	19768	19768
V8	1.0	1.0	0.0	1.0	1.0	7787	7787
V9	1.0	1.0	0.0	1.0	1.0	11628	11628
V10	0.0	0.4639	0.4639	-0.6959	1.1598	3229	3229
V11	0.0	0.479	0.479	-0.7185	1.1975	6472	6472
V12	0.0	1.0	1.0	-1.5	2.5	120	3415
V13	0.0	1.0	1.0	-1.5	2.5	410	410
V14	1.0	1.0	0.0	1.0	1.0	257	257
V15	0.0	0.0	0.0	0.0	0.0	62392	418
V16	0.0	0.0	0.0	0.0	0.0	62415	666
V17	0.0	0.0	0.0	0.0	0.0	66644	1400

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Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	IQR Outliers	Z-Score Outliers
V18	0.0	0.0	0.0	0.0	0.0	66675	1786
V19	0.8164	1.0	0.1836	0.5409	1.2754	108801	391
V20	0.8478	1.0	0.1522	0.6196	1.2282	112537	1318
V21	0.0	0.0	0.0	0.0	0.0	66247	410
V22	0.0	0.0	0.0	0.0	0.0	66282	948
V23	1.0	1.0	0.0	1.0	1.0	15703	15703
V24	1.0	1.0	0.0	1.0	1.0	25989	25989
V25	1.0	1.0	0.0	1.0	1.0	17226	17226
V26	1.0	1.0	0.0	1.0	1.0	19348	19348
V27	0.0	0.0	0.0	0.0	0.0	391	391
V28	0.0	0.0	0.0	0.0	0.0	394	394
V29	0.0	1.0	1.0	-1.5	2.5	410	5000
V30	0.0	1.0	1.0	-1.5	2.5	1632	11316
V31	0.0	0.0	0.0	0.0	0.0	71422	693
V32	0.0	0.0	0.0	0.0	0.0	71449	1142
V33	0.0	0.0	0.0	0.0	0.0	66698	431
V34	0.0	0.0	0.0	0.0	0.0	70478	702
V35	0.0	1.0	1.0	-1.5	2.5	89	3633
V36	0.0	1.0	1.0	-1.5	2.5	507	8814
V37	1.0	1.0	0.0	1.0	1.0	26343	7565

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Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	IQR Outliers	Z-Score Outliers
V38	1.0	1.0	0.0	1.0	1.0	39478	5464
V39	0.0	0.0	0.0	0.0	0.0	63577	3656
V40	0.0	0.0	0.0	0.0	0.0	63734	6261
V41	1.0	1.0	0.0	1.0	1.0	308	308
V42	0.0	0.0	0.0	0.0	0.0	63132	2331
V43	0.0	0.0	0.0	0.0	0.0	63310	6381
V44	1.0	1.0	0.0	1.0	1.0	23918	4043
V45	1.0	1.0	0.0	1.0	1.0	34544	6007
V46	1.0	1.0	0.0	1.0	1.0	9324	9324
V47	1.0	1.0	0.0	1.0	1.0	15052	15052
V48	0.0	1.0	1.0	-1.5	2.5	294	3979
V49	0.0	1.0	1.0	-1.5	2.5	956	8560
V50	0.0	0.0	0.0	0.0	0.0	69089	272
V51	0.0	0.0	0.0	0.0	0.0	68049	3274
V52	0.0	0.0	0.0	0.0	0.0	70334	5304
V53	0.0	1.0	1.0	-1.5	2.5	156	156
V54	0.0	1.0	1.0	-1.5	2.5	696	696
V55	1.0	1.0	0.0	1.0	1.0	24205	5561
V56	1.0	1.0	0.0	1.0	1.0	40744	9678
V57	0.0	0.0	0.0	0.0	0.0	63637	1967

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Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	IQR Outliers	Z-Score Outliers
V58	0.0	0.0	0.0	0.0	0.0	63844	3182
V59	0.0	0.0	0.0	0.0	0.0	64808	2932
V60	0.0	0.0	0.0	0.0	0.0	65075	5486
V61	0.8298	1.0	0.1702	0.5745	1.2553	109261	815
V62	0.8676	1.0	0.1324	0.6689	1.1987	114930	2386
V63	0.0	0.0	0.0	0.0	0.0	64483	2252
V64	0.0	0.0	0.0	0.0	0.0	64782	6235
V65	1.0	1.0	0.0	1.0	1.0	173	173
V66	1.0	1.0	0.0	1.0	1.0	99761	22665
V67	1.0	1.0	0.0	1.0	1.0	26987	26987
V68	0.0	0.0	0.0	0.0	0.0	266	266
V69	0.0	1.0	1.0	-1.5	2.5	493	5482
V70	0.0	1.0	1.0	-1.5	2.5	1727	11465
V71	0.0	0.0	0.0	0.0	0.0	69623	2188
V72	0.0	0.0	0.0	0.0	0.0	69833	3469
V73	0.0	0.0	0.0	0.0	0.0	68658	2844
V74	0.0	0.0	0.0	0.0	0.0	72566	4750
V75	0.0	1.0	1.0	-1.5	2.5	151	3827
V76	0.0	1.0	1.0	-1.5	2.5	695	695
V77	1.0	1.0	0.0	1.0	1.0	26942	7416

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Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	IQR Outliers	Z-Score Outliers
V78	1.0	1.0	0.0	1.0	1.0	43461	5391
V79	0.0	0.0	0.0	0.0	0.0	64574	2823
V80	0.0	0.0	0.0	0.0	0.0	66644	3276
V81	0.0	0.0	0.0	0.0	0.0	66723	5894
V82	0.8446	1.0	0.1554	0.6115	1.2331	98763	719
V83	0.882	1.0	0.118	0.7049	1.1771	104756	2318
V84	0.0	0.0	0.0	0.0	0.0	66130	2282
V85	0.0	0.0	0.0	0.0	0.0	66273	6708
V86	1.0	1.0	0.0	1.0	1.0	24465	4211
V87	1.0	1.0	0.0	1.0	1.0	37668	6142
V88	1.0	1.0	0.0	1.0	1.0	378	378
V89	0.0	0.0	0.0	0.0	0.0	422	422
V90	0.0	1.0	1.0	-1.5	2.5	456	5397
V91	0.0	1.0	1.0	-1.5	2.5	1650	12119
V92	0.0	0.0	0.0	0.0	0.0	72677	2166
V93	0.0	0.0	0.0	0.0	0.0	72682	3486
V94	0.0	0.0	0.0	0.0	0.0	68688	4
V95	0.0	0.0	0.0	0.0	0.0	79683	1159
V96	0.0	1.0	1.0	-1.5	2.5	81404	1298
V97	0.0	0.0	0.0	0.0	0.0	136012	1310

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Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	IQR Outliers	Z-Score Outliers
V98	0.0	0.0	0.0	0.0	0.0	31604	31604
V99	0.0	1.0	1.0	-1.5	2.5	59674	6740
V100	0.0	0.0	0.0	0.0	0.0	93702	7139
V101	0.0	0.0	0.0	0.0	0.0	34011	1136
V102	0.0	0.0	0.0	0.0	0.0	54264	1265
V103	0.0	0.0	0.0	0.0	0.0	42482	1281
V104	0.0	0.0	0.0	0.0	0.0	26699	3235
V105	0.0	0.0	0.0	0.0	0.0	42475	2009
V106	0.0	0.0	0.0	0.0	0.0	32152	1963
V107	1.0	1.0	0.0	1.0	1.0	248	248
V108	1.0	1.0	0.0	1.0	1.0	2984	2984
V109	1.0	1.0	0.0	1.0	1.0	8523	8523
V110	1.0	1.0	0.0	1.0	1.0	4665	4665
V111	1.0	1.0	0.0	1.0	1.0	1727	1727
V112	1.0	1.0	0.0	1.0	1.0	2993	2993
V113	1.0	1.0	0.0	1.0	1.0	2058	2058
V114	1.0	1.0	0.0	1.0	1.0	5538	5538
V115	1.0	1.0	0.0	1.0	1.0	18104	18104
V116	1.0	1.0	0.0	1.0	1.0	9034	9034
V117	1.0	1.0	0.0	1.0	1.0	727	727

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Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	IQR Outliers	Z-Score Outliers
V118	1.0	1.0	0.0	1.0	1.0	988	988
V119	1.0	1.0	0.0	1.0	1.0	776	776
V120	1.0	1.0	0.0	1.0	1.0	1008	1008
V121	1.0	1.0	0.0	1.0	1.0	2632	2632
V122	1.0	1.0	0.0	1.0	1.0	1380	1380
V123	1.0	1.0	0.0	1.0	1.0	15514	15514
V124	1.0	1.0	0.0	1.0	1.0	44180	7342
V125	1.0	1.0	0.0	1.0	1.0	24651	24651
V126	0.0	0.0	0.0	0.0	0.0	79683	1327
V127	0.0	107.95	107.95	-161.925	269.875	76732	1523
V128	0.0	0.0	0.0	0.0	0.0	136012	1446
V129	0.0	0.0	0.0	0.0	0.0	31604	2526
V130	0.0	59.0	59.0	-88.5	147.5	92231	9704
V131	0.0	0.0	0.0	0.0	0.0	93702	7668
V132	0.0	0.0	0.0	0.0	0.0	34011	1191
V133	0.0	0.0	0.0	0.0	0.0	54264	1402
V134	0.0	0.0	0.0	0.0	0.0	42482	1336
V135	0.0	0.0	0.0	0.0	0.0	26699	3068
V136	0.0	0.0	0.0	0.0	0.0	42475	3177
V137	0.0	0.0	0.0	0.0	0.0	32152	3072

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Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	IQR Outliers	Z-Score Outliers
V279	0.0	0.0	0.0	0.0	0.0	98358	1158
V280	0.0	1.0	1.0	-1.5	2.5	44311	1549
V281	0.0	0.0	0.0	0.0	0.0	28070	11174
V282	0.0	1.0	1.0	-1.5	2.5	22467	10224
V283	0.0	1.0	1.0	-1.5	2.5	41400	8549
V284	0.0	0.0	0.0	0.0	0.0	44976	5629
V285	0.0	1.0	1.0	-1.5	2.5	79429	5806
V286	0.0	0.0	0.0	0.0	0.0	17238	17238
V287	0.0	0.0	0.0	0.0	0.0	123664	9135
V288	0.0	0.0	0.0	0.0	0.0	99630	7976
V289	0.0	0.0	0.0	0.0	0.0	102198	7331
V290	1.0	1.0	0.0	1.0	1.0	38662	3477
V291	1.0	1.0	0.0	1.0	1.0	127598	252
V292	1.0	1.0	0.0	1.0	1.0	68377	371
V293	0.0	0.0	0.0	0.0	0.0	44085	1139
V294	0.0	0.0	0.0	0.0	0.0	71114	1491
V295	0.0	0.0	0.0	0.0	0.0	55453	1521
V296	0.0	0.0	0.0	0.0	0.0	52583	2513
V297	0.0	0.0	0.0	0.0	0.0	27788	8390
V298	0.0	0.0	0.0	0.0	0.0	47635	2417

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Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	IQR Outliers	Z-Score Outliers
V299	0.0	0.0	0.0	0.0	0.0	34899	2198
V300	0.0	0.0	0.0	0.0	0.0	19749	19749
V301	0.0	0.0	0.0	0.0	0.0	21862	5570
V302	0.0	0.0	0.0	0.0	0.0	140337	6265
V303	0.0	0.0	0.0	0.0	0.0	141029	5159
V304	0.0	0.0	0.0	0.0	0.0	140698	10156
V305	1.0	1.0	0.0	1.0	1.0	4	4
V306	0.0	0.0	0.0	0.0	0.0	98358	1348
V307	0.0	151.2917	151.2917	-226.9376	378.2293	71893	1854
V308	0.0	35.9658	35.9658	-53.9487	89.9145	105499	1726
V309	0.0	0.0	0.0	0.0	0.0	44976	2975
V310	0.0	107.95	107.95	-161.925	269.875	68650	10376
V311	0.0	0.0	0.0	0.0	0.0	17238	1439
V312	0.0	0.0	0.0	0.0	0.0	123664	8314
V313	0.0	0.0	0.0	0.0	0.0	99630	7667
V314	0.0	0.0	0.0	0.0	0.0	115998	9819
V315	0.0	0.0	0.0	0.0	0.0	102198	8000
V316	0.0	0.0	0.0	0.0	0.0	44085	1224
V317	0.0	0.0	0.0	0.0	0.0	71114	1736
V318	0.0	0.0	0.0	0.0	0.0	55453	1630

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Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	IQR Outliers	Z-Score Outliers
V319	0.0	0.0	0.0	0.0	0.0	27788	2943
V320	0.0	0.0	0.0	0.0	0.0	47635	3501
V321	0.0	0.0	0.0	0.0	0.0	34899	3216
ProductCD_W	0.0	1.0	1.0	-1.5	2.5	0	0
ProductCD_H	0.0	0.0	0.0	0.0	0.0	33024	33024
ProductCD_C	0.0	0.0	0.0	0.0	0.0	68519	0
ProductCD_S	0.0	0.0	0.0	0.0	0.0	11628	11628
ProductCD_R	0.0	0.0	0.0	0.0	0.0	37699	37699
card4_discover	0.0	0.0	0.0	0.0	0.0	6651	6651
card4_mastercard	0.0	1.0	1.0	-1.5	2.5	0	0
card4_visa	0.0	1.0	1.0	-1.5	2.5	0	0
card4_american express	0.0	0.0	0.0	0.0	0.0	8328	8328
M4_M2	0.0	0.0	0.0	0.0	0.0	59865	0
M4_M0	1.0	1.0	0.0	1.0	1.0	112691	0
M4_M1	0.0	0.0	0.0	0.0	0.0	52826	52826